

OTT CUSTOMER CHURN

ANALYSIS REPORT

Python · SQL · EDA · Power BI

Industry: Media & Entertainment | Domain: Customer Analytics

Dataset: 3,000 subscribers | 17 features | 10 OTT Platforms

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Feature Engineering

EDA

SQL Analytics

Data Cleaning

Churn Modeling

Power BI

This report presents a comprehensive analysis of customer churn across India's top OTT platforms. The findings are based on end-to-end data analytics — from raw data cleaning and exploratory analysis in Python to SQL-powered business queries and Power BI dashboarding.

1. Business Context & Problem Statement

India's OTT (Over-The-Top) streaming market has seen explosive growth, with 10+ platforms competing for subscriber attention — from Netflix India and Amazon Prime Video to Zee5, Hotstar, JioCinema, SonyLIV, MX Player, Voot, Aha, and ShemarooMe. In such a hyper-competitive landscape, **subscriber retention is not optional — it is existential**.

Acquiring a new customer costs 5–7x more than retaining an existing one. Despite this, platforms struggle to identify *which customers are about to leave, why they leave, and what can be done before they do*. The absence of

a data-driven churn prediction and segmentation framework means retention budgets are spent reactively — on customers who have already decided to leave.

Primary Business Problem: How can an OTT platform proactively identify high-risk churners, understand the behavioral and demographic drivers of churn, and deploy targeted interventions — before revenue walks out the door?

This project was designed to answer that question through rigorous data analytics across three layers: Python-based data preprocessing and EDA, SQL-driven business metric extraction, and Power BI visualization for decision-maker consumption.

2. Analytical Approach & Methodology

Dataset Overview

Total Records	Features	Platforms Covered	Cities Covered
3,000	17	10	20+
Subscribers	Variables	OTT Platforms	Indian Cities

The dataset contained 17 attributes spanning customer demographics (age group, city), subscription behaviour (plan, tenure, monthly charge), platform engagement (avg watch hours/week, number of profiles, favourite genre, primary device), friction signals (login failures, payment failures, support tickets), satisfaction (NPS score), and the binary churn outcome.

Phase 1 — Data Ingestion & Cleaning (Python / Pandas)

Raw data was loaded from an Excel file (OTT_Churn.xlsx) into a Pandas DataFrame. The following cleaning operations were applied:

- **Missing Value Imputation:** city filled with mode; monthly_charge_inr and nps_score filled with their respective medians — preserving distribution without distortion by outliers.
- **Outlier Correction:** avg_watch_hrs_week contained negative values (data entry errors). Absolute value transformation was applied to recover valid engagement data.
- **Duplicate Removal:** ott.drop_duplicates() was applied to ensure row-level uniqueness.
- **Column Standardisation:** All column names lowercased using .str.lower() for consistent downstream querying.
- **Encoding:** Target variable churned mapped from Yes/No to 1/0 binary encoding for numerical analysis and ML readiness.

Phase 2 — Exploratory Data Analysis (EDA)

- Churn distribution via countplot (class imbalance check)
- Churn by device type — countplot with hue
- Churn by payment method — countplot with hue
- NPS Score vs Churn — boxplot to capture satisfaction divergence
- Churn by Age Group — demographic segmentation
- Correlation analysis on numerical features against churn flag
- Boxplot on monthly_charge_inr for outlier detection

Phase 3 — SQL Business Analytics (MySQL)

Cleaned data was pushed to a MySQL database (ott_churn_db) via SQLAlchemy's create_engine. Eight SQL queries were written to answer specific business questions:

SQL Query	Logic	Business Output
Total Customers	SELECT COUNT(*) FROM ott_customer_churn	3,000 subscribers
Overall Churn Rate	AVG(churned) * 100	22.47%
Churn by Platform	GROUP BY platform ORDER BY churn_rate DESC	ZEE5 highest at ~31%
Churn by Subscription Plan	GROUP BY subscription_plan	Premium+ highest at 28.26%
Avg Watch Hours	ROUND(AVG(avg_watch_hrs_week), 2)	Engagement baseline
Revenue by Plan	SUM(monthly_charge_inr) GROUP BY plan	Premium plan #1 revenue driver
Top Cities	GROUP BY city ORDER BY total_customers DESC LIMIT 10	Pune leads with 384
DB Setup	CREATE DATABASE ott_churn_db	Foundation query

Phase 4 — Power BI Dashboard

Cleaned CSV was loaded into Power BI Desktop to build an executive-level dashboard. Visualisations included churn rate KPI cards, platform-wise and plan-wise churn bar charts, geographic subscriber maps, and NPS distribution slicers — enabling non-technical stakeholders to self-serve insights.

Concepts & Techniques Applied

Concept / Technique	Where Applied
Data Pipeline (ETL)	Excel → Pandas → MySQL → Power BI
Feature Engineering	abs() on watch hours; binary encoding of churn; column normalisation
Imputation Strategy	Mode for categorical (city); Median for skewed numerics (charge, NPS)
Outlier Detection	Boxplot on monthly_charge_inr; abs() fix on watch hours
EDA (Univariate & Bivariate)	countplots, boxplots, groupby aggregations across all features
SQL Aggregation & Grouping	GROUP BY, AVG(), SUM(), COUNT(), ORDER BY, LIMIT
Database Integration	SQLAlchemy ORM — Pandas → MySQL via to_sql()
Business Intelligence	Power BI dashboard with KPIs, slicers, bar & map visuals
Churn Rate Computation	AVG(binary_flag) × 100 — clean mathematical approach
Segment Analysis	Multi-dimensional slicing: platform, plan, age, device, payment, city

3. Key Findings & Data Insights

Overall Churn Rate	Monthly Revenue at Risk	Avg NPS — Churned	Watch Hrs Gap
22.47%	INR 1,96,943	3.64 / 10	1.92 hrs/wk
674 of 3,000 customers	Lost from churned users	vs 5.33 for retained	Churned watch less

3.1 Platform-Level Churn

Platform	Churn Rate	Observation
ZEE5	31.58%	HIGHEST — data quality flag (case duplicates)
MX Player	26.15%	High churn — engagement & content issues
Hotstar	25.08%	Premium pricing sensitivity
JioCinema	23.21%	Near average
SonyLIV	22.90%	Near average
Netflix India	19.66%	Below average — content stickiness
Prime Video India	20.00%	Below average — ecosystem lock-in
ShemarooMe	18.98%	LOWEST major platform churn

Note: ZEE5 appears in both title-case and uppcase (ZEE5 vs Zee5), inflating its churn to 31.58%. After normalisation, effective churn is ~25%, still among the highest. A data governance step is recommended.

3.2 Subscription Plan Churn

Subscription Plan	Churn Rate	Insight
Premium+	28.26%	Highest among paid plans — value perception gap
Free	27.40%	Expected — no switching cost
Club & Cinema	26.88%	Bundle fatigue
Super	25.64%	Mid-tier dissatisfaction
Basic	14.10%	Lowest — price-sensitive but sticky
Monthly	17.02%	Short-term commitment, lower churn
Annual	20.00%	Long-tenure — lower churn
MX Gold	17.04%	Well-positioned value plan

Revenue analysis from SQL confirms Premium plan drives the highest monthly revenue (INR 1,89,017), followed by Annual (INR 1,44,700) and Premium 4K (INR 1,38,542). Protecting these cohorts is a direct revenue imperative.

3.3 Demographic & Behavioural Segmentation

Age Group	Churn Rate	Observation
Age 26-35	26.38%	Core earning demographic — highest churn risk
Age 36-45	25.46%	Second highest — mid-career life transitions
Age 18-25	20.77%	Younger users — platform-hop behaviour
Age 46-55	19.70%	More settled viewing habits
Age 55+	19.94%	Lowest churn — loyal niche audience

Primary Device	Churn Rate	Note
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Tablet	23.47%	Highest device-based churn
Smart TV	23.31%	Living room viewer — competitive with cable
Laptop	22.67%	
Fire Stick	22.12%	
Mobile	20.70%	Lowest — most accessible, habitual use

Payment Method	Churn Rate	Note
Debit Card	24.39%	Highest — auto-renewal friction
UPI	23.01%	
Credit Card	22.58%	
Wallet	21.94%	
Net Banking	20.67%	Lowest — deliberate, sticky payment mode

3.4 Engagement & Friction Metrics

Avg Watch Hrs — Active	Avg Watch Hrs — Churned	Avg Tenure — Active	Payment Failures — Churned
13.02 hrs/wk	11.10 hrs/wk	25.35 months	2.36 avg
Retained users	~15% lower engagement	vs 22.21 churned	vs 1.90 active

Login failures also diverge: churned customers averaged 3.18 vs 2.93 for retained users. Notably, support ticket counts were nearly identical (2.41 vs 2.44) — suggesting support interactions do not strongly differentiate churn in this dataset.

3.5 Geographic Distribution (SQL Insight)

City	Subscribers	Approx Churn Rate
Pune	384	27.34%
Jaipur	244	~22%
Nagpur	242	~22%
Surat	237	~22%
Delhi	232	~22%
Bengaluru	229	~22%
Lucknow	228	~22%
Ahmedabad	222	~22%
Hyderabad	222	~22%
Chennai	222	~22%

Pune dominates subscriber count with 384 customers — 28% higher than the next city. Metro cities (Delhi, Bengaluru, Hyderabad, Chennai) each hover around 222–232 subscribers, indicating broad Tier-1 market penetration.

4. Business Impact of This Analysis

Revenue at Risk Quantified: 674 churned customers represent INR 1,96,943 in monthly revenue loss — equivalent to INR 23.6 Lakhs annually at current churn velocity. With a total monthly base of INR 9,02,481, churn is consuming **21.8% of revenue potential**.

This analysis transforms churn from a lagging metric (noticed after customers leave) into a leading signal — enabling proactive intervention. The key measured impacts are:

- ✓ **Churn Segmentation:** Platform, plan, age, device, and payment method segments are now rankable by risk level — enabling precision targeting instead of blanket retention campaigns.
- ✓ **Revenue-Weighted Prioritisation:** Premium, Annual, and Premium 4K subscribers generate the top 3 revenue streams. These cohorts need dedicated retention programs, not generic ones.
- ✓ **Engagement as Early Warning:** Churned users watch ~1.92 fewer hours/week and have 3+ months less tenure. A watch-hour drop threshold can trigger automated re-engagement.
- ✓ **Payment Friction Signal:** Payment failures (2.36 vs 1.90) and login failures (3.18 vs 2.93) are measurable pre-churn signals that can be monitored in real-time.
- ✓ **NPS as Satisfaction Barometer:** Churned users score 3.64 NPS vs 5.33 — a 31.7% gap. Low NPS users can be flagged 30–60 days before likely churn for intervention.
- ✓ **Data Infrastructure Delivered:** End-to-end pipeline (Excel → Python → MySQL → Power BI) means these insights are replicable, refreshable, and dashboard-ready — not one-time.

5. Business Recommendations

Based on the analytical findings, the following data-backed recommendations are proposed for implementation across retention, product, and technology teams:

R1	Deploy a Watch-Hour Early Warning System Set automated alerts when a subscriber's weekly watch time drops below 8 hrs/week for two consecutive weeks (vs platform average of 12–13 hrs). Trigger a personalised content recommendation push or a targeted discount offer. Expected impact: 15–20% reduction in churn among at-risk active users.
R2	Tier-Specific Retention Budgets Allocate disproportionately more retention spend on Premium, Annual, and Premium 4K subscribers — who collectively contribute INR 4,72,259/month (52% of total revenue). Even a 5% churn reduction in this segment saves INR ~23,600/month or INR 2.83 Lakhs/year.
R3	Address Premium+ Plan Value Perception Premium+ has the highest paid-plan churn at 28.26%. Conduct a price-value audit: add exclusive content, early access features, or loyalty perks to justify the premium. Consider a mid-tier 'Premium+ Lite' option to reduce downgrade-to-free behaviour.
R4	Fix Platform Data Quality (ZEE5 Case Normalisation) Standardise platform name casing in the data pipeline to avoid inflated churn metrics. A single platform appearing as 'ZEE5', 'Zee5', and other variants skews analytics. Implement a data validation layer upstream.

R5	NPS-Triggered Intervention Workflow Users scoring NPS <= 4 should enter a customer success workflow within 48 hours — personalised outreach, issue resolution, or content curation. Given the 31.7% NPS gap between churned and retained users, this is a high-ROI retention lever.
R6	Payment Failure Recovery Automation Build an automated retry + communication workflow for customers with 2+ payment failures. Offer a 7-day grace period with in-app notification, SMS, and email nudges — reducing involuntary churn from billing friction (currently showing as 2.36 avg failures in churners).
R7	26–45 Age Cohort Retention Focus The 26–35 (26.38%) and 36–45 (25.46%) segments are the highest churn-risk demographics — and also the highest-earning, highest-spending audience. Content diversification (web series, original films, family plans) and price bundling strategies should be tailored to this cohort.
R8	Tablet & Smart TV UX Optimisation Tablet (23.47%) and Smart TV (23.31%) users churn most by device. Investing in app UX, content discovery, and remote-friendly interfaces can improve stickiness. Mobile users churn least (20.70%) — mirror mobile UX patterns in other device experiences.

6. Projected Business Impact if Recommendations Applied

If the above recommendations are implemented with conservative effectiveness estimates, the projected improvements over a 12-month horizon are:

Metric	Current → Target	Change	Est. Revenue Impact
Churn Rate Reduction	22.47% → ~16–18%	~4–6% absolute reduction	INR 54,000–90,000/month saved
Premium Segment Retention	77% → 85%+ retained	+8% premium retention	INR ~15,000–25,000/month
NPS Improvement	4.95 avg → 6.0+	+21% satisfaction uplift	Reduced passive churn
Payment Recovery	2.36 failures → <1.5	Fewer involuntary churns	INR 10,000–20,000/month
Annual Revenue Impact	—	Combined recommendations	INR 7–12 Lakhs/year uplift

Bottom Line: A structured retention program powered by this analytical framework — combining watch-hour monitoring, NPS triggers, payment recovery automation, and segment-specific offers — can conservatively recover INR 7–12 Lakhs annually from what is currently being lost to preventable churn. The data infrastructure is already in place.